Homework Assignment 12: Intrinsic Mitigation

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1 General Instructions

The goal of this assignment is to familiarize you with intrinsic mitigation techniques for fairness. You will implement the techniques covered in class on the UCI Adult Census dataset. You will run two different Jupyter notebooks that have been written for you. For each, you will answer questions that demonstrate your understanding of the concepts involved. Make sure that you review the code carefully and understand how it works. Even though you are not writing code in this assignment, you will be expected to apply some of the techniques to a new dataset during the final assignment.

The total assignment is worth 200 points.

Begin by downloading the Jupyter notebooks and dataset into the same directory on your local machine. The files are:

- 1_logreg_fairness.ipynb
- 2_adversarial_debiasing.ipynb
- adult.data

Create a new conda environment from the YAML file for this homework assignment.

2 Logistic Regression with Fairness Constraints

Run the first notebook, **1_logreg_fairness.ipynb** and answer the questions below. **(9 points each)**

1. What are the categorical variables that are included in the model?
2. What is done to transform these variables for modeling?
3. What are the numeric variables in the model?
4. How are these variables transformed?
5. What three fairness metrics are calculated by *check_fairness()*?
6. Which two fairness metrics are clearly violated in the unconstrained logistic regression model? Demgraphic Parity, Error-rate parity Which one is satisfied?
7. Why does one metric, error-rate parity, differ strongly from another, predictive parity?
8. How does the constrained model differ from the unconstrained model?

9. Describe how the fairness metrics of the constrained model changed.

10. Why did Predictive Parity decrease when the other metrics improved?

11. Ignoring fairness requirements, which model performed better? What metric was calculated to measure this performance?

3 Adversarial Debiasing

Run the second notebook, **2_adversarial_debiasing.ipynb** and answer the questions below. **(9 points each)**

1. Pytorch is a Python package used for neural network programs. It manipulates data structures called *tensors*, which are multidimensional containers for numbers. In this code, the pandas dataframes are converted to tensors. What are the dimensions of the tensors *X_train_t* and *y_train_t*? (Hint: if you can't tell from the setup of the network, you can write a line of code to get the dimensions).

2. What are the layers of the predictor network?

3. How many neurons are in each layer?

4. What is the activation function used in the predictor? The final output of the predictor network is a probability. What is the function that transforms the linear result in the final layer to a probability?

5. What is the loss function used in the predictor?

6. How does the structure of the adversary network differ from that of the predictor?

7. Which fairness metric is satisfied in the baseline model?

8. How do the fairness metrics change as a result of running the adversarial network?

9. Are the AUROCs for the two models within 5%?

4 Comparison

Compare the two methods for classifying income while intrinsically mitigating for fairness. Which do you think does a better job and why? **(20 points)**